Hinglish Sarcasm Detector

Detecting sarcasm in code mixed Hindi and English text using fine tuned transformers

Subject	Natural Language Processing (ICT, Semester 7)
Institution	Pandit Deendayal Energy University, Gandhinagar
Team	Member 1: _____ Member 2: _____
NLP tasks covered	Text Classification, Sentiment Analysis

1. Project Summary

Sarcasm is one of the hardest things for a machine to understand because the words say one thing and the speaker means the opposite. The problem gets harder in India, where people mix Hindi and English in the same sentence and write Hindi in Roman letters (Hinglish). Most existing sentiment tools are built for clean English and fail on this kind of text.

In this project we build a system that reads a Hinglish comment and predicts whether it is **sarcastic** or **sincere**, along with a confidence score. We create our own labeled dataset, compare a classical baseline against modern transformer models, explain the predictions, and deliver a working web app with a public demo link.

Example inputs and outputs

Input comment	Expected label	Why
wah kya baat hai, bilkul time pe aaye	Sarcastic	Praise words used to mock lateness
thanks yaar, time pe aaye, maza aa gaya	Sincere	Genuine thanks, positive tone matches words
great, another Monday meeting, just what I wanted	Sarcastic	Positive words with a negative situation

2. What We Will Do

The project runs in eight phases over roughly four weeks.

Phase	Work done	Output
1. Planning	Fix the labels, write an annotation guideline, set up GitHub and a shared Colab notebook	Guideline document, repo
2. Data collection	Collect Hinglish comments from YouTube, Reddit and a Google Form shared with classmates, then clean spam and duplicates	3,000 to 5,000 raw comments
3. Labeling	Both members label a common set of 300 comments, measure agreement with Cohen's kappa, then label the rest	Labeled CSV dataset
4. Preprocessing	Normalize text, keep emojis and punctuation, split into train, validation and test sets	Clean dataset splits
5. Baseline model	TF IDF features with Logistic Regression to set a reference score	Baseline accuracy and F1

Phase	Work done	Output
6. Fine tuning	Fine tune MuRIL, XLM RoBERTa and HingBERT, and compare them	Best trained model
7. Evaluation	Confusion matrix, precision, recall, macro F1, error analysis and LIME explanations	Results and insights
8. App and report	Streamlit web app, deployment on Hugging Face Spaces, README and presentation	Live demo and final report

3. Dataset Plan

Source	What we take from it	Role in the project
YouTube comments	Comments from Indian comedy, roast and meme videos using the YouTube Data API	Main raw text source
Reddit	Posts and comments from Indian humor communities using PRAW	Main raw text source
Google Form	Sentences written or submitted by friends and classmates	Natural, diverse examples
SemEval 2020 SentiMix	Hinglish tweets labeled with sentiment	Pretraining step so the model learns Hinglish sentiment first
LLM generated text	A small set of sarcastic and sincere Hinglish lines, all checked by hand	Under 20 percent of data, only to fill gaps

Labeling quality: we write one strict guideline, label a shared set independently, and report Cohen's kappa as proof that the labels are reliable. Comments where we disagree are discussed and either fixed or removed.

4. Technologies, Reasons and Where We Use Them

Technology	Why we use it	Where we use it
Python	Standard language for NLP and machine learning with the best library support	Every stage of the project
pandas and NumPy	Fast handling of tabular text data and numeric results	Cleaning, splitting and analyzing the dataset
YouTube Data API and PRAW	Legal, structured access to public comments and posts	Phase 2, data collection
Google Forms and Label Studio	Easy way to crowdsource sentences and to label them with a clear interface	Phases 2 and 3, collection and labeling
emoji library	Converts emojis to text so the tokenizer can understand them, since emojis often signal sarcasm	Phase 4, preprocessing
scikit learn	Provides TF IDF, Logistic Regression and evaluation metrics in a few lines	Phase 5 baseline and Phase 7 metrics
Hugging Face Transformers and Datasets	Pretrained models and a ready Trainer for fine tuning, the same workflow we used in our BERT experiment	Phase 6, model training
PyTorch	Deep learning engine that runs underneath the Transformers library	Phase 6, training and inference
MuRIL	Pretrained on Indian languages including Roman script text, so it suits Hinglish	Main model
XLM RoBERTa	Strong multilingual model that gives a fair comparison point	Comparison model
HingBERT	Pretrained specifically on Hinglish text	Comparison model
LIME	Shows which words pushed a prediction toward sarcastic, making the model explainable	Phase 7, explanations and app display
matplotlib and seaborn	Clear charts and confusion matrices for the report	Phase 7, evaluation
Google Colab or Kaggle	Free GPU, enough for short comments and small models	Phases 5 and 6, training
Streamlit	Builds an interactive web interface in pure Python with no frontend work	Phase 8, user interface
Hugging Face Spaces	Free hosting that gives a public demo link to share with recruiters and faculty	Phase 8, deployment
GitHub	Version control, teamwork and a public portfolio page	All phases, plus final README

Why transformers and not only classical methods

- TF IDF treats words independently, so it misses the mismatch between positive words and a negative situation that defines sarcasm.
- Transformers read the full sentence and capture context, word order and tone.
- We keep the TF IDF model as a baseline so we can show exactly how much the transformer improves.

Preprocessing decisions specific to sarcasm

- Keep emojis and punctuation, because signals like "!!!" and "..." carry tone.
- Do not remove stop words, because small words such as "bilkul" and "kya" shape sarcastic meaning.
- Normalize stretched words, for example "waaaah" becomes "waah", to reduce noise.
- Use stratified splits so every class appears in the same proportion in train, validation and test data.

5. How We Will Evaluate

Measure	What it tells us
Accuracy	Overall share of correct predictions
Precision and recall	How trustworthy the sarcastic predictions are, and how many real sarcastic comments we catch
Macro F1	Balanced score across classes, our main metric because sarcasm data is usually imbalanced
Confusion matrix	Which kinds of mistakes the model makes
Cohen's kappa	How much the two of us agree when labeling, which shows dataset quality
Error analysis	A manual review of about 30 wrong predictions to find patterns and limits

6. Expected Outcomes

Deliverables

- A labeled Hinglish sarcasm dataset of about 3,000 to 5,000 comments with a documented guideline and agreement score.
- A trained and compared set of models: a TF IDF baseline, MuRIL, XLM RoBERTa and HingBERT.
- A Streamlit web app where a user types a comment and sees the label, a confidence score and the words that influenced the result.
- A "try to fool the model" mode and a feedback button that stores corrections for future retraining.
- A public demo on Hugging Face Spaces, a complete GitHub repository with README, and a final report with a short presentation.

Target results

Model	Target macro F1	Purpose
TF IDF with Logistic Regression	About 0.65 to 0.72	Baseline reference
Fine tuned MuRIL or HingBERT	0.80 or higher	Main result
XLM RoBERTa	Close to the best model	Comparison

These are goals based on similar published work on code mixed text. Final numbers will depend on our data size and label quality, and we will report the real results honestly.

Learning outcomes

- Hands on experience with the complete NLP pipeline: data collection, annotation, modeling, evaluation and deployment.
- Practical skill in fine tuning transformer models and comparing them against a baseline.
- Understanding of code mixed language, a real and active research challenge.

- Experience with explainable AI, so we can justify why the model made a decision.

Value for our profile

- A live demo link and a clean GitHub project that can be shown directly to recruiters.
- A self built dataset, which proves data skills that most student projects lack.
- Measurable results that can be written on a resume, with real numbers in place of the targets above.
- Relevant for roles in applied AI, NLP engineering, social media analytics, customer experience and content moderation.

7. Possible Risks and Our Plan

Risk	How we handle it
Sarcasm is subjective and labels may disagree	Strict guideline, shared labeling set, and removing comments we cannot agree on
Small dataset	Pretrain on SentiMix sentiment first and use light augmentation
API limits or blocked scraping	Rely more on Google Form submissions and public datasets
Overlap with a roast style class	Start with two classes and add a third class only if time allows